



Ain Shams University

Faculty of Engineering · Mechatronics & Robotics Department

Thermal Management of a Battery-Powered Medical Payload Pod

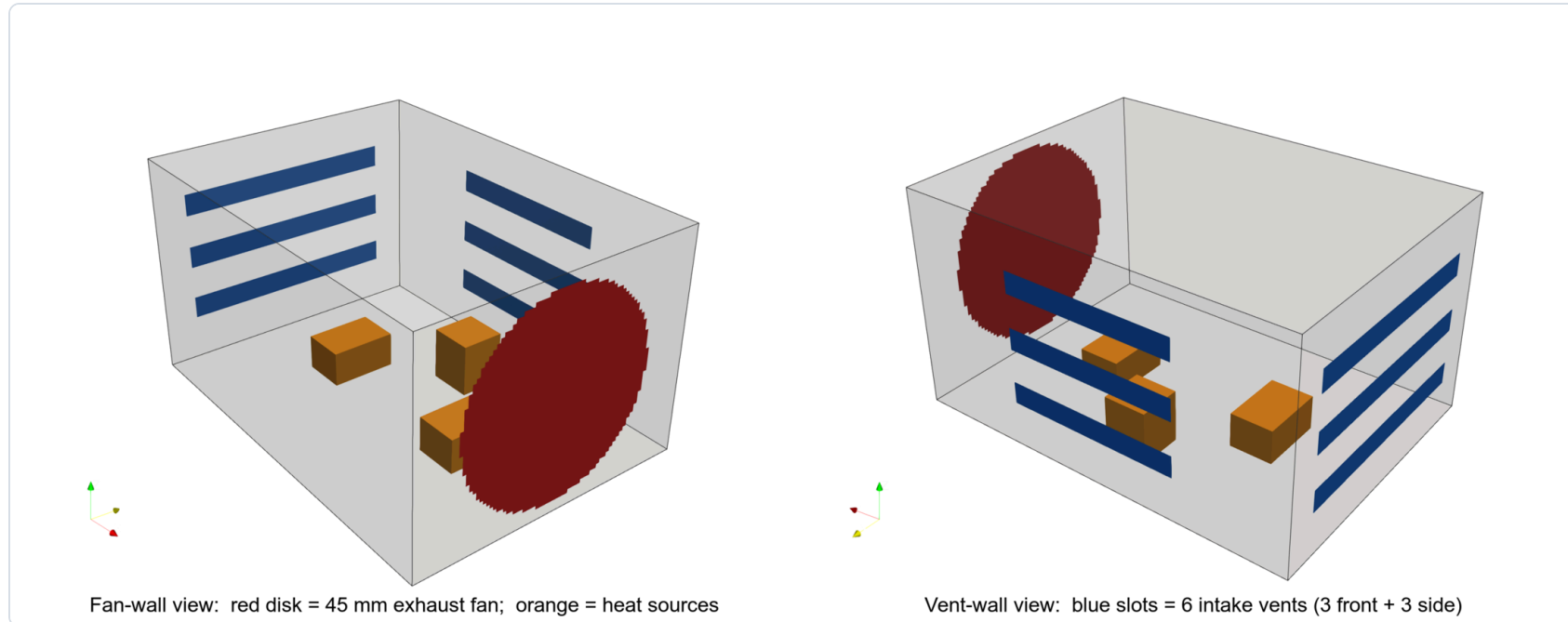
A scaled prototype study inspired by drone-delivered emergency medical care

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Heat Transfer · MEP 311s · Presented to Dr. Amr Elbanhawey · 16 June 2026

The Problem

- ◆ Drone-delivered medical payload pod (Zipline-style): **three** thermally managed enclosures — power, control, plant
- ◆ The **plant module** dissipates **~17 W** (two 8 Ω resistors + IRF530 MOSFET)
- ◆ **Constraint:** every module must stay $\leq 40\text{ }^{\circ}\text{C}$ in steady state
- ◆ Judged on thermal performance **and** cost-to-weight ratio

Design — Forced-Air Fan + Vents



- ◆ One **50 mm 5 V exhaust fan** (45 mm opening) + **six intake slots**, in a $90 \times 65 \times 50$ mm PLA box
- ◆ Selected over a **fin array** ($\approx 0.30 \text{ m}^2$ of fins, far heavier & bulkier)

Analytical Thermal Analysis

- ◆ **Power:** $P = V^2/R = 8^2/4 = \mathbf{16\ W}$ resistive ($\sim 17\ \text{W}$ total)
- ◆ **Fin array — rejected:** needs $\sim 0.30\ \text{m}^2$ over two faces; heavy and hard to mount
- ◆ **Forced-air energy balance:** $\Delta T = Q/(\rho \cdot \dot{V} \cdot c_p) \approx \mathbf{4.6\ ^\circ\text{C}}$ $\rightarrow$ cavity $\sim 30\text{--}35\ ^\circ\text{C}$
- ◆ Bracketed by the sealed-box bound ($118\ ^\circ\text{C}$); confirmed by CFD & experiment

CFD Simulation BONUS

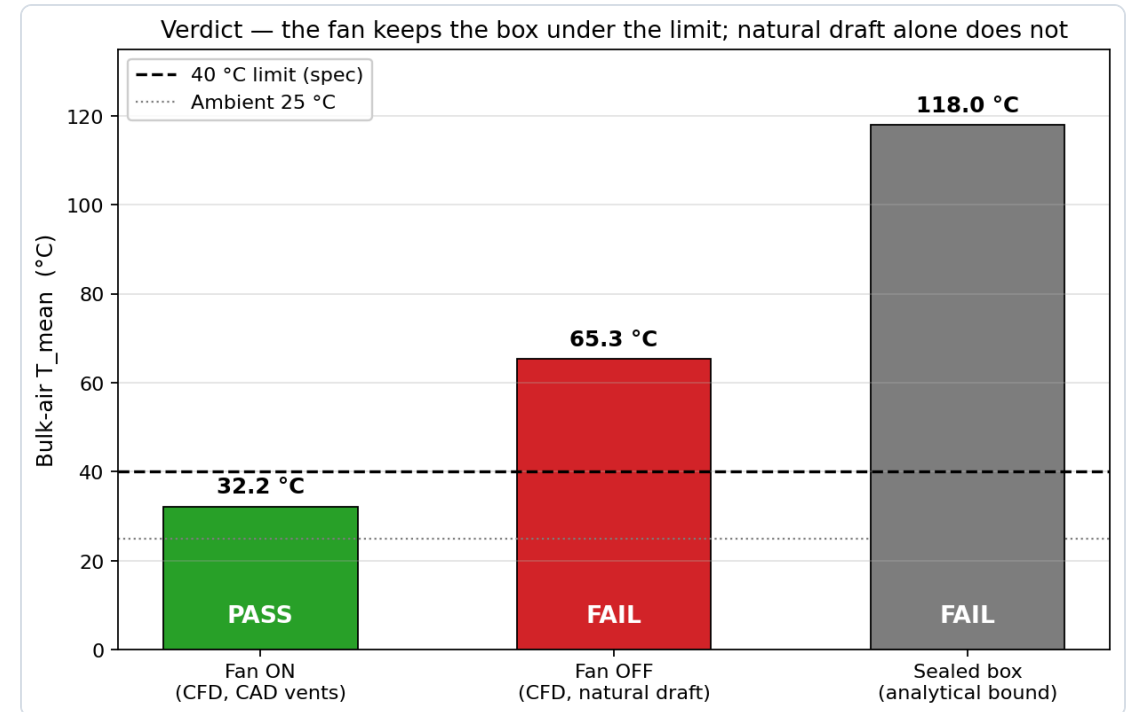
- ◆ OpenFOAM · k- ω SST · **528 k cells**
- ◆ **Fan ON → 32 °C → PASS**
- ◆ Fan OFF → 65 °C → FAIL
- ◆ Heat hugs the floor; $\sim\frac{3}{4}$ leaves as warm fan exhaust — energy balance closes on 17 W

32 °C

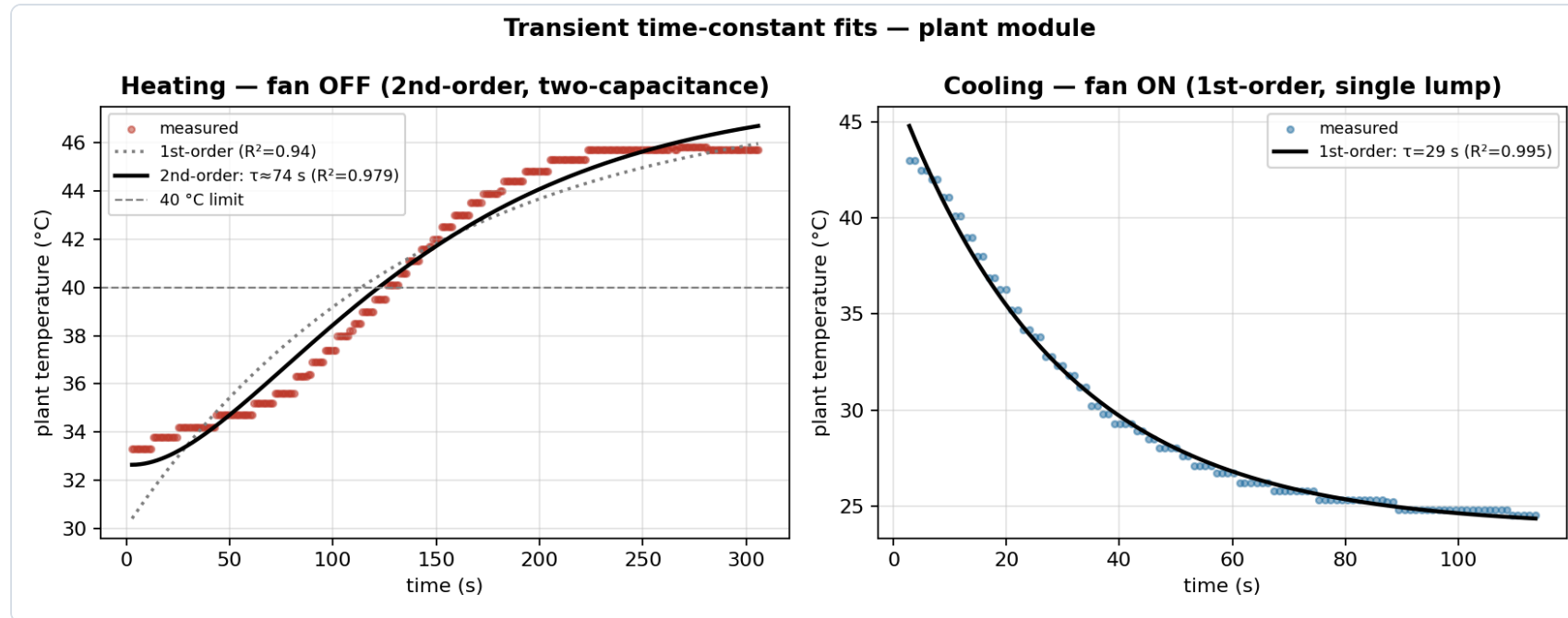
Fan ON (PASS)

65 °C

Fan OFF (FAIL)

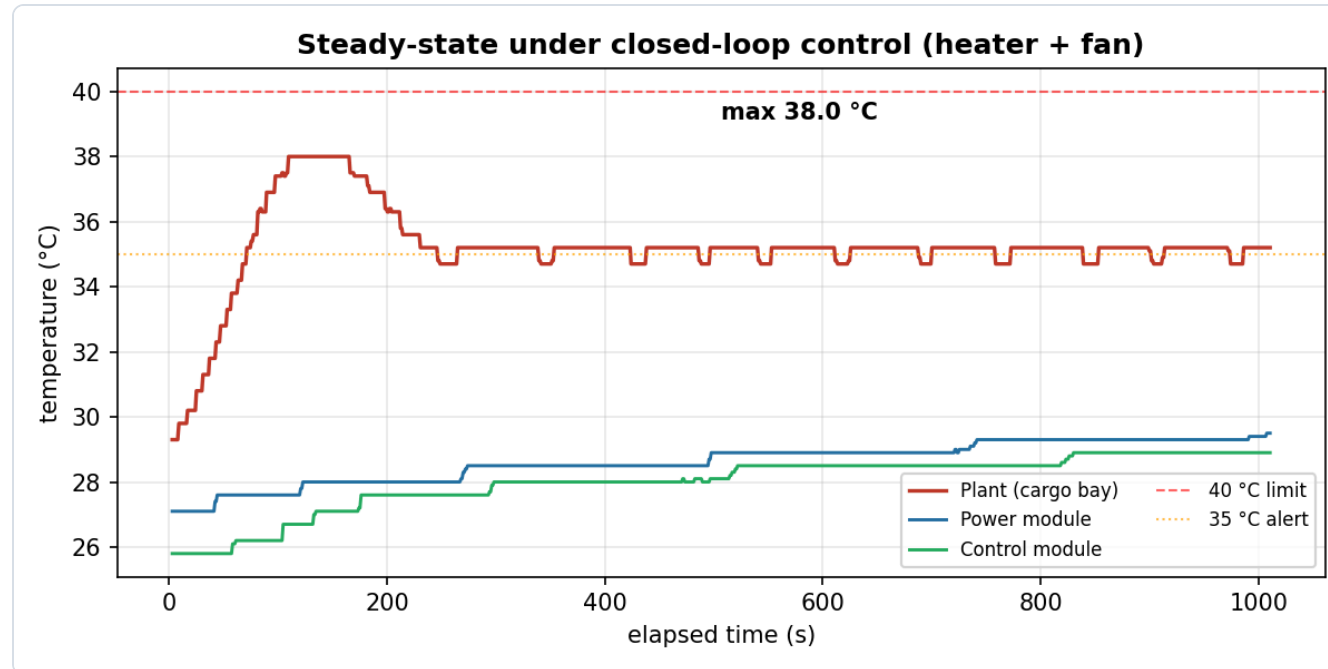


Transient Response



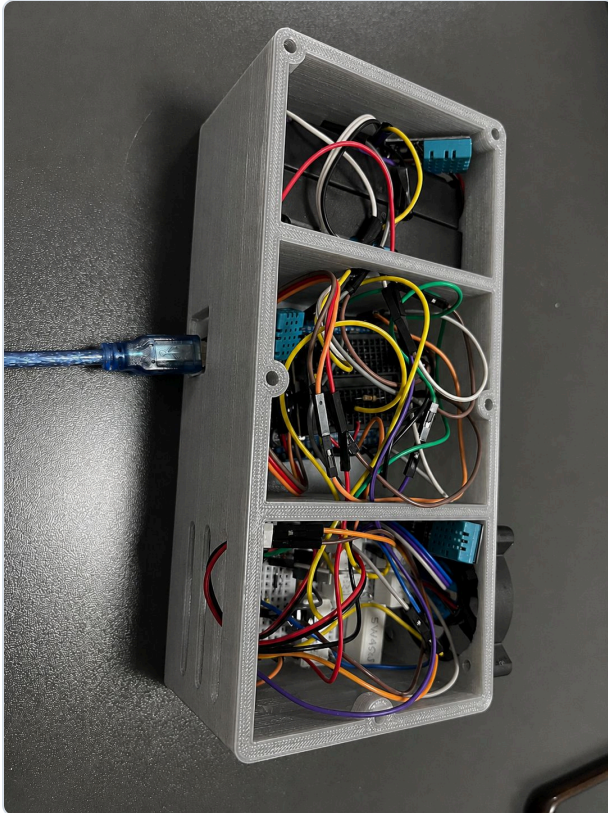
- ♦ **Cooling (fan on):** clean **first-order**, $\tau = 29$ s ($R^2 = 0.995$); predicted $C/UA \approx 20\text{--}30$ s
- ♦ **Heating (fan off):** **second-order** — two cascaded thermal capacitances (S-shaped)

Experimental Results



- ◆ Closed-loop control holds the plant at **~35 °C** (< 40 °C); power & control ~29 °C
- ◆ Measured full-heat draw **17.1 W** — matches the 17 W used in the analysis

Prototype & Cost

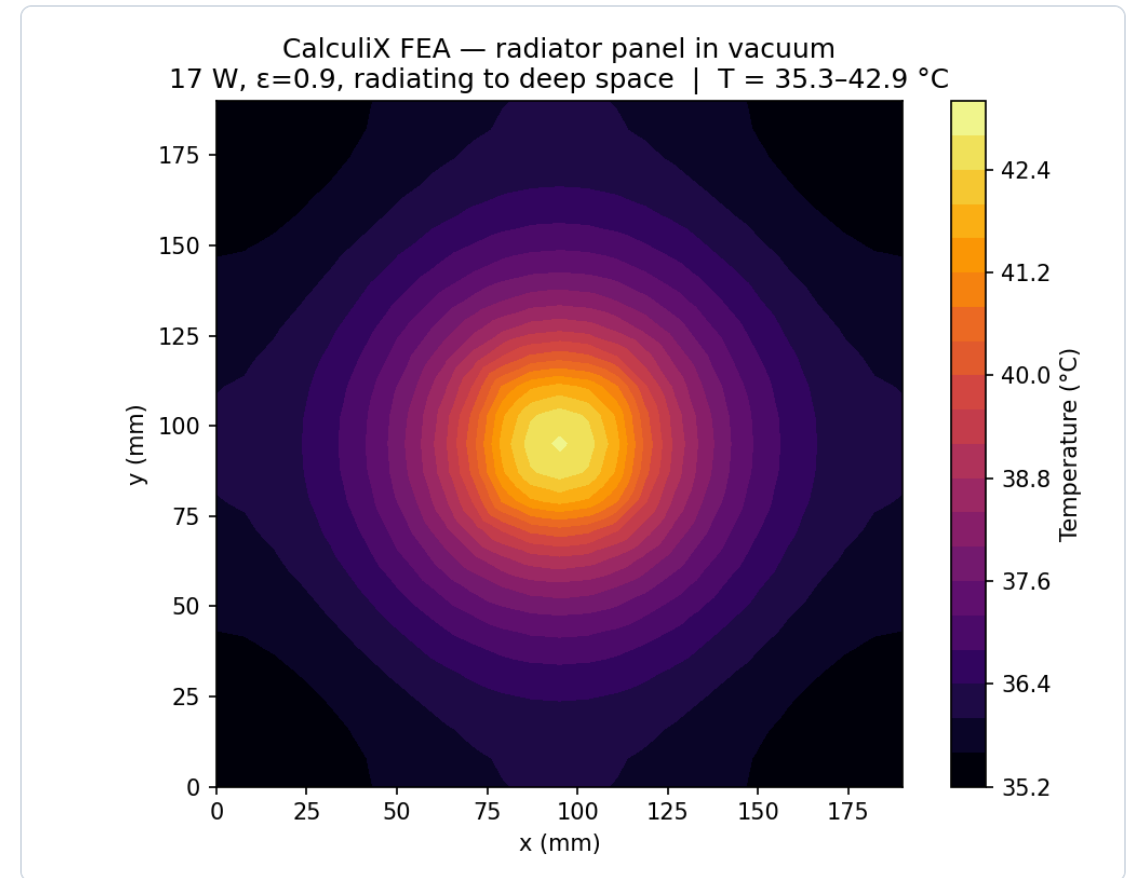


- ◆ 3-D-printed three-compartment PLA pod
- ◆ Arduino **PWM** control of the MOSFET; 3× DHT sensors; graded LED alert + 40 °C cutoff
- ◆ Battery + load **current/power** monitored in firmware
- ◆ Thermal-management cost-to-weight: **80 EGP / 55 g $\approx$ 1.45 EGP/g**

Space-Environment Variant

BONUS

- ◆ In vacuum there is **no convection** — the fan/vents do nothing
- ◆ Heat is rejected by **radiation** to deep space via a high-emissivity panel
- ◆ Sizing: a **~0.035 m²** radiator holds the module at 40 °C
- ◆ **CalculiX FEA:** panel 35–43 °C (mean ~37 °C) — confirms the sizing



Conclusion

- ◆ **All modules $\leq 40\text{ }^{\circ}\text{C}$** in steady state — constraint met and experimentally confirmed
- ◆ The fan is mandatory (removes $33\text{ }^{\circ}\text{C}$); the vents are a passive safety net
- ◆ Two analytical methods + CFD + experiment all agree ($\sim 32\text{--}37\text{ }^{\circ}\text{C}$)
- ◆ Bonuses delivered: **numerical** (CFD + CalculiX FEA) and the **space** variant
- ◆ Code & data: [**github.com/omarkeshk/medical-payload-pod-thermal**](https://github.com/omarkeshk/medical-payload-pod-thermal)